

Table 48. I/O current injection susceptibility

Symbol	Description		Functional susceptibility ⁽¹⁾		Unit
			Negative injection	Positive injection	
I _{INJ}	Injected current on pin	Any IO	5	NA	mA

1. Evaluated by characterization. Not tested in production.

5.3.13 I/O port characteristics

General input/output characteristics

Unless otherwise specified, the parameters given in [Table 49](#) are derived from tests performed under the conditions summarized in [Table 23: General operating conditions](#). All I/Os are designed as CMOS- and TTL-compliant.

For information on GPIO configuration, refer to the application note AN4899 *STM32 GPIO configuration for hardware settings and low-power consumption*, available on the ST website www.st.com.

Table 49. I/O static characteristics

Symbol	Parameter	Conditions		Min	Typ	Max	Unit
V _{IL} ⁽¹⁾	I/O input low level voltage	All	2.0 V < V _{DDIO1} < 3.6 V	-	-	0.3 × V _{DDIO1}	V
V _{IH} ⁽¹⁾	I/O input high level voltage	All	2.0 V < V _{DDIO1} < 3.6 V	0.7 × V _{DDIO1}	-	-	V
V _{hys} ⁽²⁾	I/O input hysteresis	-		-	200	-	mV
I _{lkg} ⁽³⁾	Input leakage current ⁽³⁾	0 < V _{IN} ≤ V _{DDIO1}		-	±70	-	nA
		V _{DDIO1} ≤ V _{IN} ≤ V _{DDIO1} + 1 V		-	600	-	
		V _{DDIO1} + 1 V ≤ V _{IN}		-	150	-	
R _{PU}	Weak pull-up equivalent resistor ⁽⁴⁾	V _{IN} = V _{SS}		25	40	55	kΩ
R _{PD}	Weak pull-down equivalent resistor ⁽⁴⁾	V _{IN} = V _{DDIO1}		25	40	55	kΩ
C _{IO}	I/O pin capacitance	-		-	5	-	pF

1. Refer to [Figure 17: I/O input characteristics](#).

2. Specified by design. Not tested in production.

3. This parameter represents the pad leakage of the I/O itself. The total product pad leakage is provided by the following formula: I_{Total_leak_max} = 10 μA + [number of I/Os where V_{IN} is applied on the pad] × I_{lkg}(Max).

4. Pull-up and pull-down resistors are designed with a true resistance in series with a switchable PMOS/NMOS. This PMOS/NMOS contribution to the series resistance is minimal (~10% order).